JANYA KANSARA

SOFTWARE ENGINEER · BACKEND & SYSTEMS · DEVELOPER TOOLS

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SUMMARY

Computer Science undergraduate with hands-on experience across the full development lifecycle, from system design through deployment, gained by independently building backend services, search infrastructure, and simulation engines. Comfortable writing clean, testable, efficient code and quick to pick up new tools and technologies. Seeking to bring this foundation to Motadata as a Software Engineer – Trainee.

EDUCATION

LJ University, Ahmedabad
B.Tech / B.E. in Computer Science & Technology

Expected 2028
CGPA: 8.75 / 10

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, TypeScript, SQL, Rust
Backend	FastAPI, Django, Express, REST APIs, SQLAlchemy, Alembic
Frontend	React, HTML, CSS, Tailwind CSS
Databases	PostgreSQL, SQLite, MySQL, MongoDB
Tools	Git, Docker, Linux
Core	Object-Oriented Programming, Data Structures & Algorithms, Concurrency, Database Design, Indexing, Caching

PROJECTS

Qwry — Self-Hostable Search Engine

React · FastAPI · Rust · Tantivy · PostgreSQL · SQLAlchemy · Valkey · SearXNG

- Built a full-stack search engine with a React interface, FastAPI backend, and a custom **Rust/Tantivy** indexing engine.
- Indexed **38K+ documents** and integrated **~70 search providers** via SearXNG for federated search.
- Achieved **~230ms average local search latency** through indexed retrieval, PostgreSQL persistence, and Valkey caching.
- Designed REST APIs and backend services for search, discovery, indexing, and workspace functionality.

DirStudio — Directory Analysis & Management Tool

Python · FastAPI · JavaScript · SQLite

- Built a directory analysis application for scanning, inspecting, and organizing large, deeply nested file structures.
- Reduced scan time on a 1.2K-file, 4.4GB test directory from 73s to 28s (**~62% faster**) via asynchronous scanning and parallel workers.
- Implemented filesystem-tree merging, duplicate detection, and metadata processing in a Python/FastAPI backend.

3D Particle Life Simulation

Java · Object-Oriented Programming · Simulation

- Developed a real-time 3D particle simulation in Java with configurable, per-species attraction and repulsion rules.
- Optimized simulation performance to sustain **~58 FPS at 5K particles** and **~24 FPS at 10K particles**.
- Applied modular OOP design separating simulation state, particle behavior, configuration, and rendering.

CERTIFICATIONS & ACTIVITIES

- Inheritance and Data Structures in Java — University of Pennsylvania, Coursera
- Exploratory Data Analysis for Machine Learning — IBM, Coursera
- Introduction to Java — LearnQuest, Coursera
- Introduction to HTML, CSS & JavaScript — IBM, Coursera
- LJ Innovation Village — Participant